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Project Title:	Creation and Transplantation of Imunoprivileged Sertoli-Neuron-Aggregated-Cells (SNACs) for the Treatment of Parkinson's Disease		
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Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
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Task Description:	Develop procedures for aggregating Sertoli Cells and immature neurons using microgravity technology, characterize the aggreagates and assess their . Nerve cell transplantation is a promising procedure for the treatment of Parkinson's Disease (PD) and other neurodegenerative diseases of the central nervous system (CNS). The focus of this research project is to create, characterize and transplant stable tissue constructs composed of Sertoli cells and immature dopaminergic neurons into a parkinsonian rat model that 1) secrete dopamine, 2) contain intrinsic trophic support mechanisms for the transplanted neurons, and 3) have inherent immunoprotective properties. Immunoprivileged cell aggregates can be transplanted into the central nervous system to treat neurodegenerative disease. These aggregates would not require systemic immunosuppression of the recipient, and would provide a modulated source of dopamine that would mimic normal brain function.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful creation of immunopriveliged neuronal aggregates that express specific neurotransmitters would provide cell replacement therapy for neurodegenerative disease and traumatic brain injury.
Task Progress:	Neuronal Differentiation of NT2 Cells Within SNAC Tissue Constructs: We have successfully achieved our goal of defining culture conditions that reliably grow SNAC tissue constructs in the HARV that contain a large number of dopaminergic cells and have determined that ratios of Sertoli-to-NT2 cells of 1:1 to 1:4 yield the greatest number of tyrosine hydroxylase (TH)-positive neurons. Ratio of Sertoli:NT2 cells The ratio of Sertoli:NT2 cells incubated in the HARV is very important to the formation of a large number of dopaminergic cells in the SNAC. We examined a series of Sertoli-to-NT2 cell ratios by counting the number of tyrosine hydroxylase (TH) – positive NT2N cells present in sections of SNAC incubated for 3 days in the HARV that were started at 1:1, 1:2, 1:4, or 1:8 Sertoli:NT2 cell ratios. These very interesting results suggest that there is a critical physical or paracrine interaction of Sertoli cells with NT2 cells that is required for the differentiation of NT2 cells. Interestingly, fetal midbrain, the source of neurons used for human fetal neuron transplants for PD, contains approximately four times as many TH+ neurons as we are able to harvest from one HARV, which suggests that it may be possible to produce a sufficient number of TH+ neurons for human transplants, should this method prove to be beneficial in animals transplant studies. While the NT2N cells in the SNAC may be TH-positive, there is still the possibility that they may not be dopaminergic. Two different strategies were adopted to determine if, in fact, the TH+ NT2N neurons seen in the HARV were dopaminergic. The most direct demonstration involved stimulation of SNAC with 30mM KCL, which stimulates the release of dopamine from dopaminergic cells, for the last hour of their incubation period in the HARV, and analysis of the amine content of the supernatant via high performance liquid chromatography with electrochemical detection. These results indicate that dopamine is released by SNAC when stimulated in the HARV. Under basal unstimulated conditions there was a low amount of DA in the SNAC medium (0.013 ± 0.01 nM). When the cells were stimulated with 30 mM KCl there was an order of magnitude increase in the DA detected in the medium (0.39 ± 0.07 nM). Neither media from SC aggregates or NT2 aggregates, nor media alone, had detectable levels of DA under basal or stimulated conditions. To confirm that the NT2N neurons contained in the SNAC are mature dopaminergic neurons, we also examined whether they contained aldehyde degydrogenase 2 (ALDH 2), an enzyme known to be found in high quantities in mature dopaminergic neurons of the substantia nigra, but not in neuronal precursors. NT2N neurons in the SNAC had a significantly higher level of ALDH 2 than did NT2 precursor cells or Sertoli cells grown under similar conditions. Differentiation of NT2 cell aggregates We have been puzzled by the consistent presence of both TH and ALDH-2 in immunoblots of NT2 aggregates, and have begun to examine whether the aggregation of NT2 cells in the HARV induces NT2 cell differentiation to NT2N neurons. Aggregation of NT2 cells, either as HARV-grown aggregates or NT2 spheres grown in suspension culture, initiates differentiation of NT2 cells to NT2N neurons, which is never seen in conventionally cultured NT2 cells. After 4 days in suspension or HARV aggregation, about 2% of NT2 cells, as compared to 9% of NT2 cells grown in SNAC, differentiate to dopaminergic NT2N neurons. This suggests that there is a common mechanism that induces differentiation by the aggregation of neuronal precursor cells, which may presage the response of stem cells in the body. Therefore, it becomes important to understand the signaling pathways involved in the initiation of differentiation. These may be signaling pathways that could be altered in microgravity, and which may be critical in maintaining the health of astronauts exposed to long-duration space travel. Possible pathways for NT2 cell differentiation: We have previously reported that the PI3/AKT pathway is, at least partially, implicated in the differentiation of NT2 cells in SNAC grown in the HARV. Inhibition of the ERK1/ERK2 pathway with the protein kinase C inhibitor GF109203X (10nM; Ki=10nM) did not inhibit the differentiation of TH+ NT2N neurons, wherease inhibition of the AKT pathway with the the phosphatidylinositol 3 inhibitor LY294002 (180nM; Ki=50µM) significantly decreased TH in the SNAC. Phosphorylated AKT was also inhibited by LY294002, but not GF109203X. We reasoned that cell adhesion molecules may also play a significant role in initiating differentiation of NT2 cells. Conventional cultures of NT2 cells express high levels of N-cadherin and connexin 43. As these NT2 cells differentiate to NT2N neurons in conventional culture in the presence of retinoic acid, both N-cadherin and connexin 43 are downregulated. When NT2 spheres grown in suspension culture and NT2 cell aggregates grown in the HARV were examined, both N-cadherin and connexin 43 were downregulated, indicating that NT2 cells grown under these culture conditions were differentiating. The Wnt pathway is involved in the differentiation of a variety of cells, and is intimately related to N-cadherin. herefore, we have started to examine the role of Wnt in the differentiation of NT2 cells. Upon binding of N-cadherin to the cell membrane, ß-catenin is released into the cytosol. If the ß-catenin becomes phosphorylated it is degraded. Unphosphorylated ß-catenin may act as an secondary signaling messenger and translocate to the nucleus. Our preliminary data indicates that GSK3ß, which phosphorylates free ß-catenin that is released into the cytosol following the binding of N-cadherin, is downregulated in NT2 spheres, but not in NT2 conventional culture. This suggests that free ß-catenin in NT2 spheres is not degraded, but allowed to act as a second messenger that translocates to the nucleus to effect cell differentiation. Based on these results we have developed a model by which NT2 cells differentiate to NT2N neurons in the absence of retinoic acid. This model suggests that increased levels of AKT1, which is known to inhibit the action of GSK3ß, may allow increased amounts of ß-catenin to remain in the cytosol, increasing the likelihood of its translocation to the nucleus to affect differentiation. An intriguing outcome of these data is that modulation of the Wnt/ß-catenin pathway and the PI3/AKT pathway, or alteration of their interaction may be significantly effected in an environment of sustained microgravity, which may alter the normal differentiation potential of developing cells or adult stem cells. Transplant of SNAC: Analysis of short term (3 month) transplant experiments to assess graft survival/rejection, and the phenotype of surviving NT2N neurons within the transplant have shown that a significant population of TH+ NT2N neurons are present in these 3 month-old transplants. This is extraordinarily important since NT2 cell differentiated with retinoic acid lose their dopaminergic phenotype after 1 week in vivo. Our findings indicate that differentiation of NT2 cells in the

	HARV, produces a stable phenotype change in the cells. Secondly, since 9% of the NT2 cells differentiate to dopaminergic neurons, we have begun to examine if some NT2 cells differentiate to other neurotransmitter phenotypes. We have found that SNAC contain both GABAergic and catecholaminergic neurons. Similarly, GABAergic neurons were seen in the three month transplants as well as catecholaminergic neurons. It is not currently known if the same ratio of dopaminergic:GABAergic:catecholaminergic neurons seen in the SNAC is retained in the transplant, or if this ratio changes in favor of one type of neurotransmitter phenotype.
Bibliography Type:	Description: (Last Updated: 06/14/2005)
Articles in Peer-reviewed Journals	Cameron, D.F; Hushen, J.J.; Colina, L.; Mallery, J.; Willing, A.E.; Sanberg, P.R.; Saporta, S. "Formation and Structure of Transplantable Tissue Constructs Generated in Simulated Microgravity from Sertoli Cells and Neuron Precursors. " Cell Transplant. 13:755-763, 2004 , Mar-2005
Articles in Peer-reviewed Journals	Saporta, S.; Willing, A.E.; Shamekh, R.; Bickford, P.; Paredes, D.; Cameron, D.F. "Rapid Differentiation of NT2 cells in Sertoli-NT2 Cell Tissue Constructs Grown in the Rotating Wall Bioreactor" Brain Res. Bull. 64:347-356, 2004. , Feb-2005
Presentation	Shamekh, R.; Cameron, D.F.; Willing, A.E.; Colina, L. and Saporta, S. "An alternative dopaminergic cell source for cell transplantation therapy in Parkinson’s Disease. " American Society for Neural Transplantation and Repair Exp. Neurol. 193:259, 2005. Apr-2005